



Cairo University
Faculty of Engineering
Credit Hours System – Senior Level
Data Mining, Big Data, and Analytics – CMPS451



Project Report

Prepared by:

Alaa Eldin Hamed	(1210017)
Ezz Eldin Ahmed Saber	(1210016)
Hussein Essam Mahmoud	(1210005)
Omar Mohamed Raouf	(1210155)

Table of Contents

Table of Contents	2
Table of Figures	3
1. Problem Description	4
2. Project Pipeline	4
3. Analysis and solution to the problem:.....	5
3.1. Data preprocessing and feature engineering:.....	5
3.2. Visualization & 3.3. Extracting insights from data:	7
4. Model/Classifier training.....	18
5. Results and Evaluation	22
6. Unsuccessful Trials	26
7. Enhancements and Future work.....	26

Table of Figures

Figure 1: Ratings Distribution	7
Figure 2 Genre popularity vs Quality	8
Figure 3 Rating distribution by Runtime	9
Figure 4 Rating Trends over Decades	10
Figure 5 Number of Titles for each language.....	11
Figure 6 Average Rating by Language	12
Figure 7 numVotes Anlaysia	13
Figure 8 Feature Correlation Heatmap.....	14
Figure 9 Director Track Record & Cast Quality vs. Title Rating	15
Figure 10 Rating Tier Distribution	16
Figure 11 User Rating Agreement.....	17
Figure 12 Model Evaluation Metrics Summary	22
Figure 13 Random Forest Regressor Feature Importances.....	24
Figure 14 Random Forest Classifier Confusion Matrix	25

1. Problem Description

Streaming platforms and media enterprises frequently acquire movies and TV series to improve viewer satisfaction and retention. However, these acquisition decisions are often based on limited manual judgment, brand reputation, or past popularity rather than a systematic predictive analysis.

The core business problem is: **How can a streaming enterprise estimate the likely audience reception of a title before acquiring it?**

2. Project Pipeline

1. Preprocessing

- Load and clean 7 IMDb TSV files and user ratings using PySpark.
- Filter, clean, and join datasets to create a unified data source.

2. Feature Engineering

- Create features such as genre one-hot encoding, runtime buckets, director/cast track records, language features, and user statistics.

3. Visualization

- Generate 10+ publication-quality plots to analyze language bias, number of votes, and other trends.

4. Modeling

- Train 4 regression models (Linear, Decision Tree, Random Forest, GBT), 3 classifiers (Logistic, Decision Tree, Random Forest), and perform K-Means clustering using Spark MLlib.

5. Evaluation

- Assess models on train/validation/test splits.
- Report metrics, confusion matrix, and feature importance.
- Provide business insights based on results.

3. Analysis and solution to the problem:

3.1. Data preprocessing and feature engineering:

After loading seven massive IMDb TSV datasets into PySpark. We start by filtering the dataset to only include movies, tv series, tv movies, tv mini-series, so that we avoid podcasts or short movies. We then remove adult content. We also drop rows missing critical values like release year, runtime, and genres. Joined the primary tables and filtered out 229,123 titles with fewer than 100 votes to ensure rating reliability. Later we integrated a 4.67 million record user ratings dataset from IEEE DataPort (IMDb Users' Ratings Dataset) and merged aggregated user features with the main dataset.

We cast the columns that are supposed to be numeric like start year, runtime, number of votes into integers. We cast ratings to float. This is done so that later we can use them as input for machine learning.

We do one-hot encoding for 27 unique genres and 4 title types, as machine learning models can not understand text, and to prevent bias.

Other Engineered features include runtime buckets (short, standard, long, very long), release decades, and a binary English language indicator.

Created a classification target by splitting the averageRating into tiers: Low (<5), Medium (5-7), and High (≥ 7)

We convert continuous values into simpler categories so models can detect patterns more easily. Movie runtimes are grouped into buckets like “short,” “standard,” “long,” and “very long” based on their length, and then each category is turned into one-hot encoded vectors. Columns with values of 0 or 1. Additionally, the movie's release year is transformed into its corresponding decade by dividing the year by 10, rounding it down, and multiplying it back, so a year like 1995 becomes 1990, this is done so the machine learning models don't get confused by the continuous numbers.

Because the language in the dataset is heavily skewed towards English, we handle this bias by adding a binary feature, if it is 1 then the title primary language is English.

We apply a transformation on the number of votes, as some movies have millions of votes and some have only hundreds. By taking the logarithm for

the number of votes. We compress the number of votes to a smooth curve. We do this to prevent right skewed data.

We then ask the question “is this director known for making good movies?” we do this by calculating the average rating for each director. If a movie has multiple directors, we create a row for each director. Afterwards we append the director’s average rating to the original data frame by joining on the primary director. If a director has no prior movies in the dataset we fill his rating with the median rating.

Next, we create a feature for cast quality. We load the principals table and drop rows with cast members other than actor and actress. We then calculate their historical track records; we do this by joining the actor list with main dataset to retrieve the average rating for every movie they appear in. by grouping the data we can generate an average rating for each actor. We then calculate the overall cast quality by averaging the scores of the top 3 billed actors together.

3.2. Visualization & 3.3. Extracting insights from data:

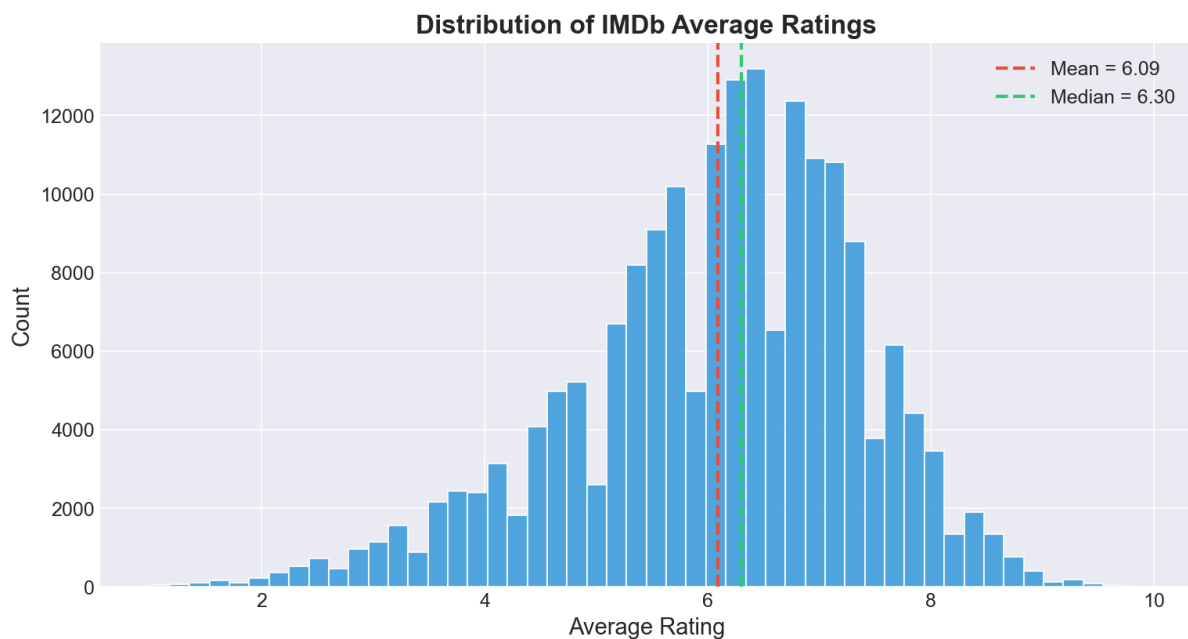


Figure 1: Ratings Distribution

Purpose: To establish a baseline understanding of the overall shape, spread, and central tendency of our primary target variable (averageRating).

Observation: The histogram displays a distinct negative (left) skew. The bulk of the data clusters heavily between the 5.0 and 7.5 marks, with the median rating (6.30) sitting noticeably higher than the mean (6.09). There is a long, thin tail stretching down toward 1.0, while ratings sharply drop off after 8.0.

Insight: Audiences are generally generous, usually giving content average or slightly above average ratings. Ratings above 8.0 are rare. Because the data is heavily concentrated in the middle, predictive algorithms will naturally bias toward predicting a 6.0 or 6.5 if not provided with strong distinguishing features.

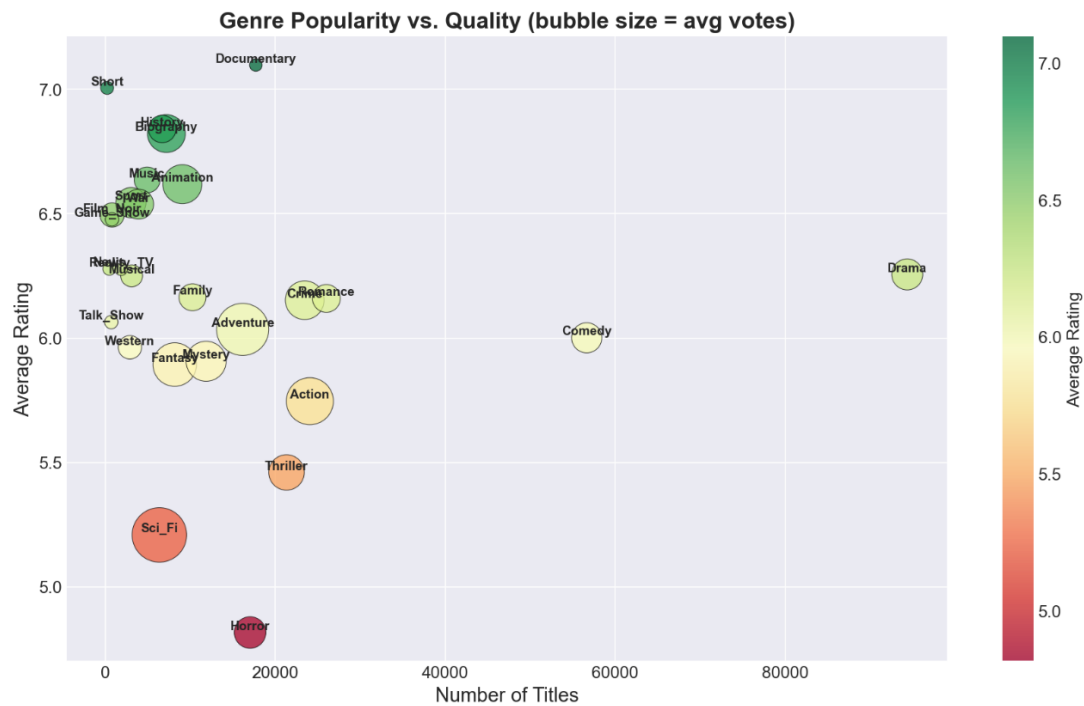


Figure 2 Genre popularity vs Quality

Purpose: To evaluate the relationship between a genre's production volume, average rating, and overall audience engagement (average votes)

Observation: The bubble chart reveals a clear separation of genres. "Drama" and "Comedy" dominate the x-axis in sheer production volume. The highest-rated genres (green bubbles in the top-left) are niche categories like "Documentary," "Biography," and "History." Conversely, "Sci-Fi," "Action," and "Adventure" feature massive bubble sizes, indicating incredibly high user engagement and vote counts, despite hovering at lower, mediocre average ratings (5.2 to 6.0). "Horror" sits isolated at the bottom with the lowest critical reception.

Insight: High ratings don't always mean big audiences. Documentaries get great scores but few views. Blockbusters like Sci-Fi and Action draw massive crowds but get harsher, lower scores. "Drama" is the perfect sweet spot, it brings in big audiences while still keeping ratings consistently high.

Action: This visually confirms that a title's genre dictates different audience grading curves. To account for this, we one-hot encoded all 27 unique genres into our feature matrix (e.g., genre_Action, genre_Drama), allowing our classification algorithms to contextualize a title's rating based on its specific genre's baseline expectations.

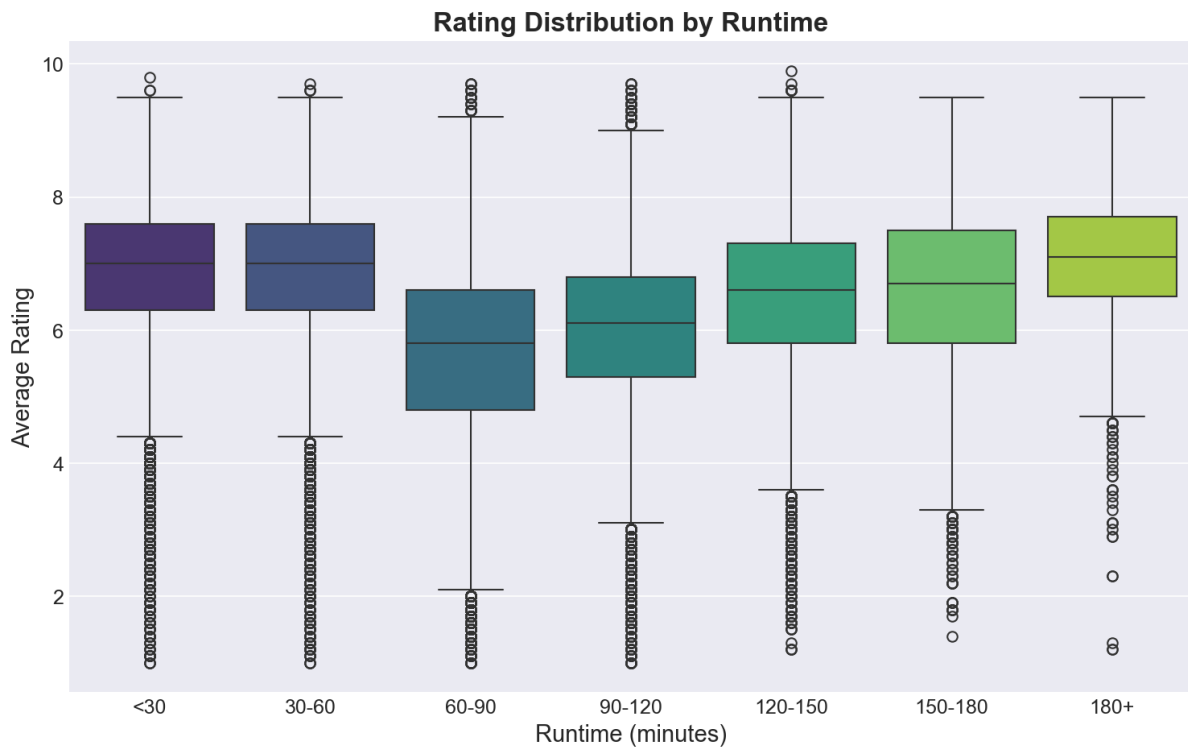


Figure 3 Rating distribution by Runtime

Purpose: To determine if the length of a title (its runtime) has any measurable impact on its average audience rating.

Observation: The boxplot visualizes the spread of average ratings across different runtime categories. Titles with runtimes under 60 minutes show a relatively high median rating (around 7.0) but have a significant number of low-rating outliers. There is a noticeable dip in median ratings for titles between 60 and 90 minutes. However, as runtimes increase beyond 90 minutes, the median rating steadily climbs, reaching its peak in the 180+ minute category.

Insight: While very short content can score well, there's a clear trend showing that longer, feature-length titles (90+ minutes) tend to receive higher average ratings. The 60-90 minute range seems to be a challenging spot for critical reception. Long-form content, particularly titles exceeding 150 minutes, appears to satisfy audiences more consistently.

Action: We engineered these runtime categories into distinct buckets (e.g., 'short', 'standard', 'long', 'very_long') to provide our predictive models with clear, categorical signals about a title's length, allowing the algorithms to factor this trend into their quality forecasts.

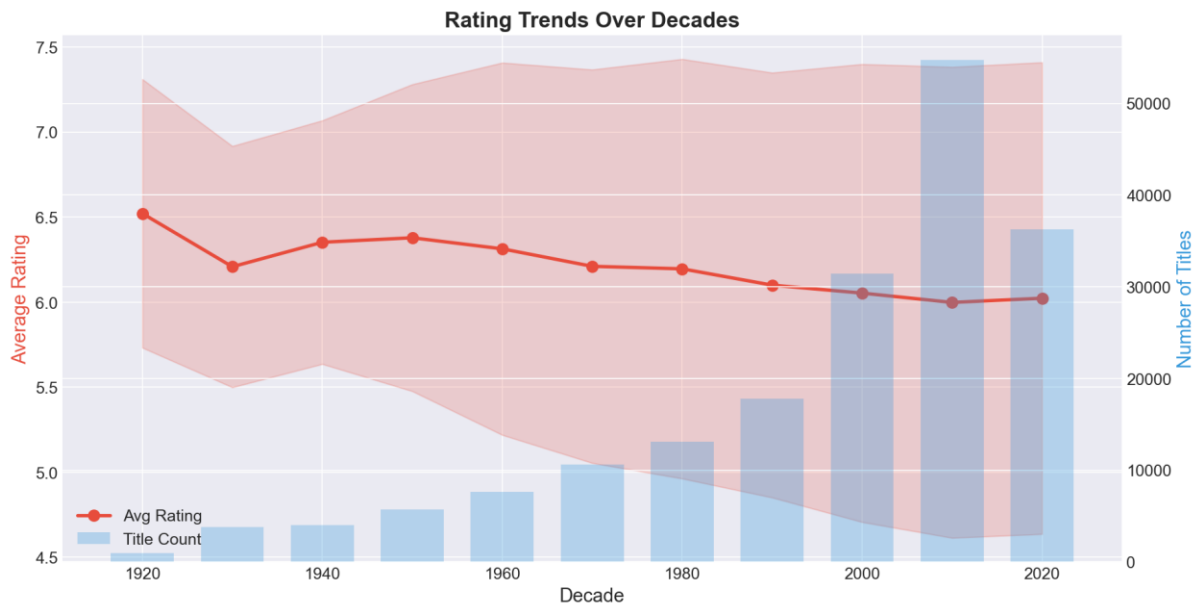


Figure 4 Rating Trends over Decades

Purpose: To see if the era a movie was made affects its average rating, especially given the massive explosion of content production in recent years.

Observation: The chart uses two axes. The blue bars show the sheer volume of movies made, which stays low for decades before skyrocketing in the 2000s and peaking in the 2010s. The red line shows the average rating, which does the exact opposite: it slowly but steadily drops from the 1940s onward, hitting its lowest point just as content volume hits its highest.

Insight: Quantity dilutes quality. The modern digital and streaming eras brought a massive flood of new movies and shows, but a lot of it is just average "filler" content. Because of this, modern titles generally have a lower average rating than older, classic films from past decades.

Action: We engineered a "decade" category and fed it into our machine learning models. This allows the algorithm to adjust its baseline expectations, knowing that a modern movie faces a lower average rating trend compared to a title from the 1970s or 80s.

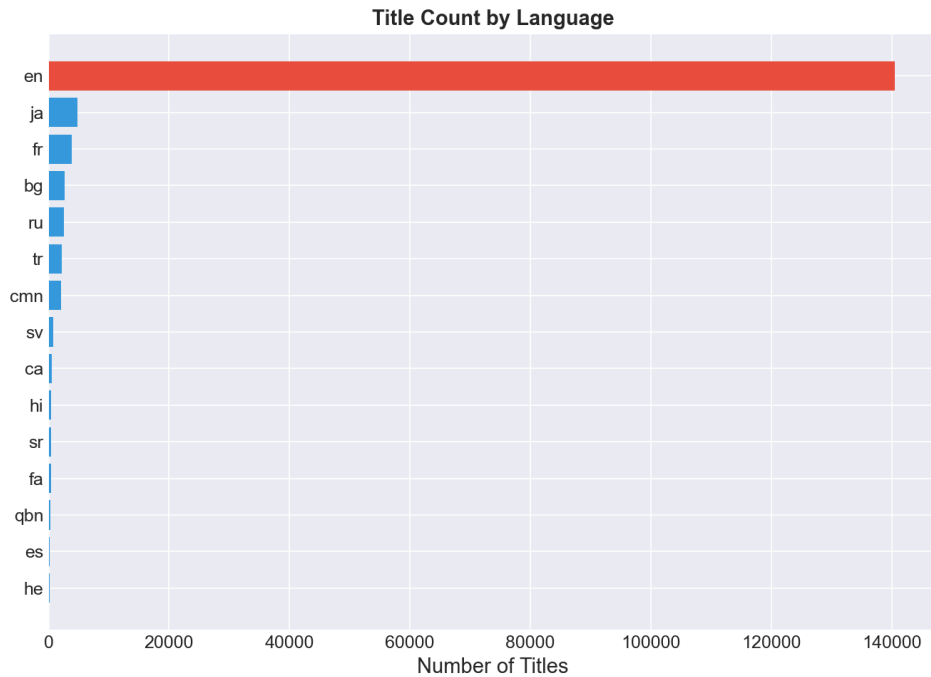


Figure 5 Number of Titles for each language

Purpose: To see if the dataset is balanced across different languages or if it leans heavily toward one specific language, which could affect our model's predictions.

Observation: The bar chart clearly shows a massive imbalance. English ("en") titles completely dominate the dataset, with over 140,000 titles. The next most common language, Japanese ("ja"), has a tiny fraction of that number, barely registering on the same scale, and all other languages have even fewer titles.

Insight: This data is heavily skewed toward English-language content. While this makes sense for a platform like IMDb, it means our predictive models will be very good at evaluating English movies but might not be as accurate or reliable when evaluating international or non-English content.

Action: We created a simple "isEnglish" feature for our models. This helps the algorithm understand that English titles are the norm and to treat non-English titles differently, acknowledging the massive difference in data availability for those categories.

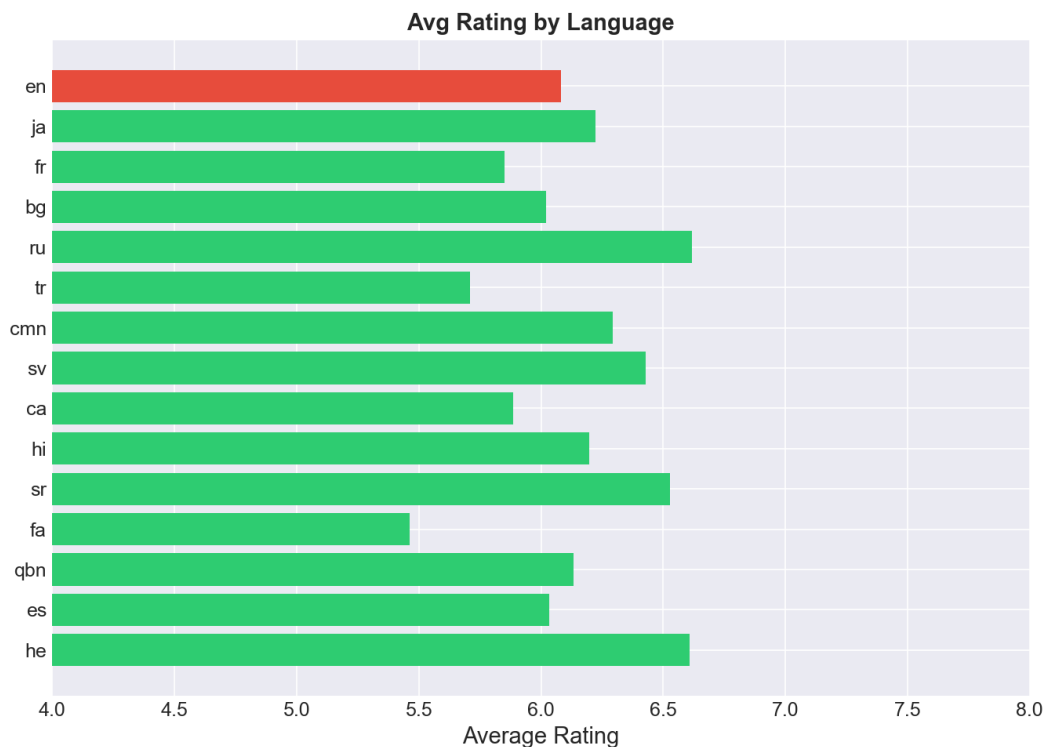


Figure 6 Average Rating by Language

Purpose: To see how the average rating changes depending on the language of the title, and to compare our dominant language (English) against international content.

Observation: The bar chart shows the average rating for the top 15 languages. Even though English ("en", highlighted in red) has the most titles by far, its average rating sits in the middle of the pack at roughly 6.1. Several international languages, like Russian ("ru"), Hebrew ("he"), and Serbian ("sr"), score noticeably higher, averaging between 6.5 and 6.6.

Insight: This highlights a "niche audience" effect. English titles are watched by the mass market, which brings in a wider, harsher range of reviews. Foreign language films, on the other hand, usually attract a smaller, more dedicated audience that actively seeks them out, leading to more generous average ratings.

Action: This perfectly reinforces why we added the "isEnglish" feature to our dataset. It helps the machine learning model understand that a 6.1 is a perfectly normal baseline for a mass-market English film, whereas the baseline expectation for an international film might naturally be skewed higher.

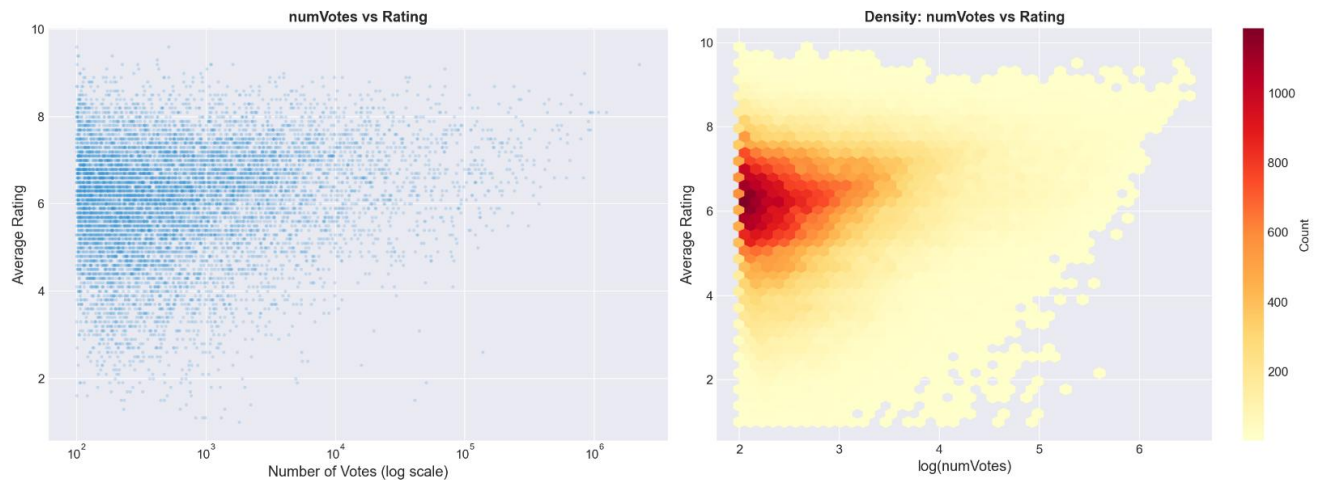


Figure 7 numVotes Analysis

Purpose: To see if the number of votes changes how reliable a movie's rating is, and to justify our decision to filter out movies with very few votes.

Observation: The charts show a clear "funnel" shape. Movies with very few votes (on the left side of the charts) have wildly different ratings, stretching all the way from 1 to 10. As the number of votes increases (moving to the right), the ratings tighten up and settle around a 6.0 to 7.0 average. The density map on the right confirms that the vast majority of our data sits in that middle-score, lower-vote area.

Insight: Movies with only a handful of votes can easily be skewed by a small group of super-fans or haters. You need a larger crowd to get a true, stable consensus of a movie's actual quality.

Action: Because of this, we set a strict cutoff rule during our data cleaning: we removed any title with fewer than 100 votes. This eliminates statistical noise and ensures our predictive models only learn from solid, audience-tested data.

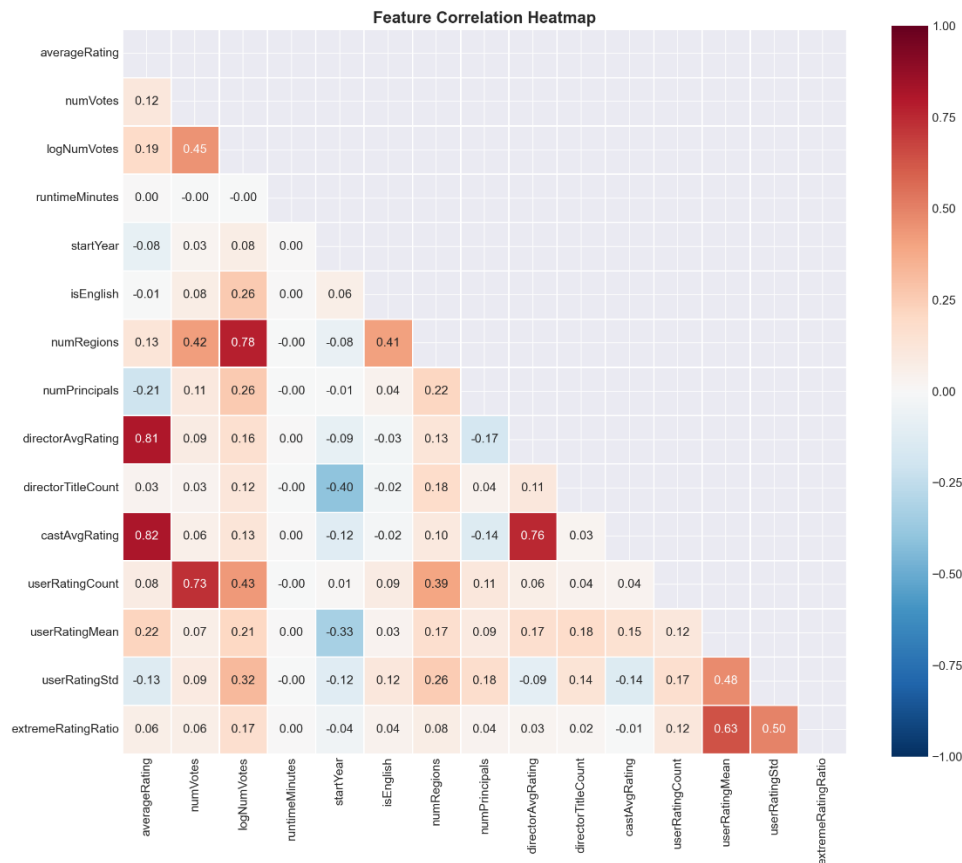


Figure 8 Feature Correlation Heatmap

Purpose: To find out which features (like runtime or director history) are most strongly linked to a movie's final rating, and to check if any features could cause multicollinearity.

Observation: The heatmap uses color to show relationships. Dark red means a strong positive link, and dark blue means a strong negative link. The most striking discovery is in the first column (averageRating). It shows an incredibly strong positive link with directorAvgRating (0.81) and castAvgRating (0.82). Meanwhile, basic details like runtimeMinutes (0.00) or isEnglish (-0.01) show almost zero connection to the final rating.

Insight: The historical track record of a director and the main cast are the best predictors of whether a new movie will get a good rating. In contrast, things like how long the movie is, or whether it's in English, don't give us much of a clue about its quality.

Action: We made sure to engineer and keep these "talent track record" features for our machine learning models, as they are clearly the strongest signals for predicting a movie's success.



Figure 9 Director Track Record & Cast Quality vs. Title Rating

Purpose: To visually check if a movie's final rating is directly tied to the historical success of its director and main cast.

Observation: Both scatter plots show a very tight, upward-sloping pattern. The red dashed trendline cuts straight through the middle of the dots in both charts, showing a near-perfect diagonal match: as a director's or cast's historical average rating goes up on the bottom axis, the specific movie's rating goes up right along with it on the side axis.

Insight: A movie's quality is heavily dependent on the people making it. If you have a highly-rated director and a highly-rated cast, you are almost guaranteed to get a highly-rated movie. Past performance is a massive and reliable indicator of future success.

Action: Because these relationships are so remarkably strong and predictable, we made sure our machine learning models heavily rely on these "talent track record" features to forecast a title's exact rating.

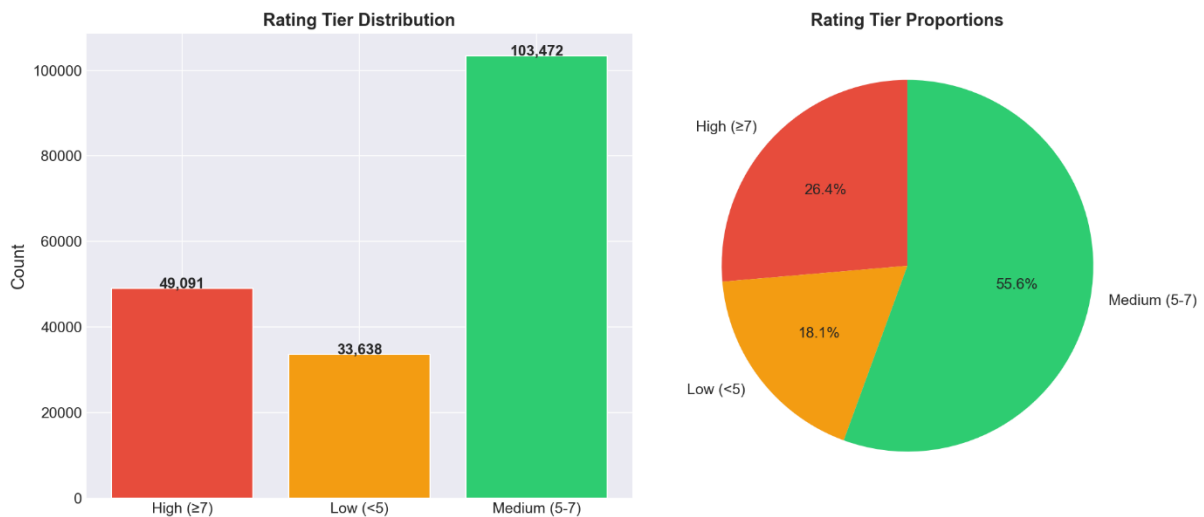


Figure 10 Rating Tier Distribution

Purpose: To see how many movies fall into our "Low", "Medium", and "High" quality buckets before we try to teach a machine to predict them.

Observation: The bar and pie charts show a massive imbalance. The "Medium (5-7)" group is huge, making up 55.6% of the entire dataset (over 103,000 titles). "High" ratings are less common at 26.4%, and "Low" ratings are the rarest at just 18.1%.

Insight: Most content in the entertainment world is simply average. Because the dataset is so unbalanced, a lazy machine learning model could just guess "Medium" for every single movie and still be right more than half the time.

Action: Knowing this, we had to change how we measure success. We couldn't just use basic "accuracy." Instead, we used the F1 Score, which penalizes the model if it just lazily guesses "Medium" all the time, ensuring it is actually learning how to spot the rare "High" and "Low" movies.

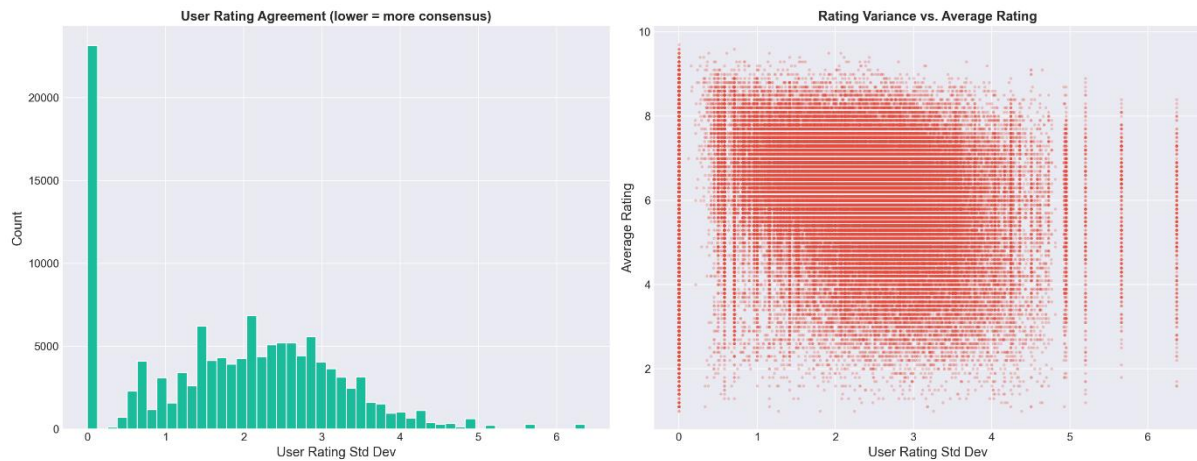


Figure 11 User Rating Agreement

Purpose: To see if viewers generally agree on a movie's quality, or if some movies loved by some but hated by others.

Observation: The histogram has a massive, unnatural spike exactly at zero. However, looking past that anomaly to the scatter plot on the right reveals the actual trend: movies with a wide, messy mix of ratings (a high standard deviation) almost always sit in the "average" 4.0 to 8.0 rating range. The absolute best (9-10) and absolute worst (1-2) movies have much tighter distributions.

Insight: That giant spike at zero caused by missing data filled with zeros. The true behavioral takeaway is that "average" movies are highly polarizing, while genuinely great or terrible movies create much stronger audience consensus.

Action: We kept `userRatingStd` (rating variance) as a feature so the algorithm can tell if a movie is universally agreed upon or highly controversial. Furthermore, spotting these artificial zeros confirms that our decision to use Tree-based models (like Random Forest) was correct, as they can easily handle and ignore these artificial zero-fill values without breaking the underlying math.

4. Model/Classifier training.

We combine all our input features into a single vector, and we add it as a column, this is done for each row. We then use StandardScaler to standardize them to have a mean of 0 and a unit variance. We then rename the target variables to fit pyspark's defaults, averageRating becomes label_reg (for regression) and ratingTier becomes label_cls (for classification).

Next, we split the data into 70% training data, 15 % validation and 15% for testing.

For K-Means we calculated the silhouette score which measures the average distance between the point and other points in its local cluster, and measures separation which is the average distance between the point and all points in the nearest neighboring cluster.

1. The "Prestige Content" Subset (K=3)

This subset grouped titles based on how good the movie is and the historical track record of the director.

Features: directorAvgRating, castAvgRating, directorTitleCount, extremeRatingRatio, decade, numPrincipals

Use case: separate creator-led prestige titles from mainstream-safe catalog.

- **Cluster 2: The Modern (104k titles):** This is the high-quality, modern catalog. They have longer runtimes (averaging 127 minutes, typical for a major theater release), high average ratings (6.68), and are made by directors with good track records (6.69).
- **Cluster 1: The Classics (37k titles):** It isolated older content. The average release year here is 1967. They have solid ratings and good directors, representing the "Golden Age" catalog.
- **Cluster 0: The "B-Movie" (44k titles):** This is the bad titles. Modern releases (2008) with terrible average ratings (4.51) made by directors who historically make bad movies (4.72 director rating).

2. The "Production Scale" Subset (K=5)

This subset grouped titles based on their runtime and vote volume, revealing the physical "format" of the content.

Features: runtimeMinutes, numRegions, numPrincipals, numRoleTypes, runtime_long, runtime_very_long

Use case: segment lean productions vs large-scale/global productions.

- **Cluster 1: The TV Series (37k titles):** The average runtime here: **55.6 minutes**. It successfully grouped 1-hour TV dramas and shows together. They have high ratings (6.71), which is typical for TV shows on IMDb.
- **Cluster 2: The Major Cinema Releases (16k titles):** These feature a 130-minute average runtime and high vote volumes (7.13 log_votes). These represent big-budget, feature-length cinema.
- **Cluster 3: The Mini-Series / Box Sets (7k titles):** The average runtime here is a massive **736 minutes** (over 12 hours). These are limited series, full TV seasons, or extended documentaries that IMDb categorizes as a single title.
- **Cluster 0 & 4: The 90-Minute Filler (95k titles):** This is the largest group. They are exactly feature-length (90 mins) but have below-average ratings (5.72). This is the standard, mediocre "filler" content that makes up the bulk of streaming platforms.

3. The "Audience Reception" Subset (K=4)

This subset grouped titles based entirely on how audiences reacted to them.

Features: numVotes, logNumVotes, userRatingMean, userRatingStd, userRatingRange, extremeRatingRatio

Use case: identify broadly-loved vs polarizing vs niche titles.

- **Cluster 0: Blockbusters (1,196 titles):** This is the best group. It is tiny (only ~1,200 titles) but has massive engagement (13.01 log_votes) and the highest overall ratings (7.45).
- **Cluster 2: The Niche Masterpieces (26k titles):** The avg_user_rating here is massive at **8.26**, yet the total vote count is low (5.95). These are long (233 mins) titles that a small number of people watched, but absolutely loved. (Likely acclaimed foreign films, niche documentaries, or theater recordings).

- **Cluster 1 & 3: The Mainstream (~158k titles):** Most titles fall here. They get average ratings (around 5.9 - 6.0) and moderate user engagement.

4. The "Temporal Trend" Subset (K=3)

This subset grouped titles based heavily on release year and the shifting trends in runtime and audience engagement over time.

Features: startYear, decade, numVotes, logNumVotes, directorTitleCount

Use case: distinguish timeless back-catalog vs recency-driven demand.

- **Cluster 0: The Contemporary Bulk (118k titles):** The largest segment represents the modern explosion of content (2011 average). Runtimes have gotten longer (122 mins), but the ratings are entirely average (5.93), showing that more content doesn't always mean better content.
- **Cluster 1: Mid-Century (41k titles):** Older titles (averaging 1965) with a tight 92-minute runtime. They hold up well with solid ratings (6.23) and represent a reliable back-catalog.
- **Cluster 2: Modern Hits (25k titles):** Recent releases (2009 average) that command massive audience attention (9.70 log_votes) and strong ratings (6.57). This is the content driving current pop culture.

5. The "Value for Money" Subset (K=6)

This subset grouped titles to find the intersection of critical quality, user satisfaction, and production cost (implied by runtime and director).

Features: runtimeMinutes, numVotes, castAvgRating, directorAvgRating, userRatingMean, numRegions, numPrincipals, extremeRatingRatio

- **Cluster 2: The Big Hits (800 titles):** Massive votes (13.25), big overall ratings at (7.60), and great directors (7.03). These are the most valuable.
- **Cluster 5: The Hidden Gems (48k titles):** Very high user satisfaction (7.74) and good overall ratings (6.71) on a standard 92-minute runtime. These are highly efficient acquisitions that audiences love but don't cost much money.

- **Cluster 1: The 90s Staples (35k titles):** Reliable, high-engagement (8.55 log_votes) catalog titles from the 90s with strong ratings. Good for platform retention.
- **Cluster 0 & 4: The Poor Investments (~101k titles):** Bad ratings (4.23 - 6.39), terrible user satisfaction (2.00 - 3.52), and poor director track records. These are low-value acquisitions that inflate a library size without adding quality.
- **Cluster 3: The Data Anomaly (1 title):** This cluster contains exactly 1 title with a runtime of 3,692,080 minutes (about 7 years long). This highlights a dirty data point in the IMDb dataset that the algorithm successfully quarantined.

6. The "Genre Format Signature" Subset (K=6)

This subset grouped titles looking at runtime formats and how specific formats historically perform.

Features: all genre columns + all type columns

- **Cluster 5: Dramas (6.7k titles):** A small group of 103-minute films with high ratings (6.82) and top directors (6.71).
- **Cluster 2: The Long-Form (57k titles):** Very long runtimes (154 mins) with average ratings (6.06). This likely captures historicals, Bollywood films, or extended director's cuts.
- **Cluster 1: The Hour-Long Formats (19k titles):** At 74 minutes, these are likely TV specials, British television formats, or short documentaries. They have good ratings (6.19).
- **Cluster 0, 3 & 4: The Standard Features (~101k titles):** Grouped around the 90 to 104-minute mark, these represent the standard movie format. Cluster 4 performs slightly better (6.54), while Cluster 3 underperforms (5.58).

5. Results and Evaluation

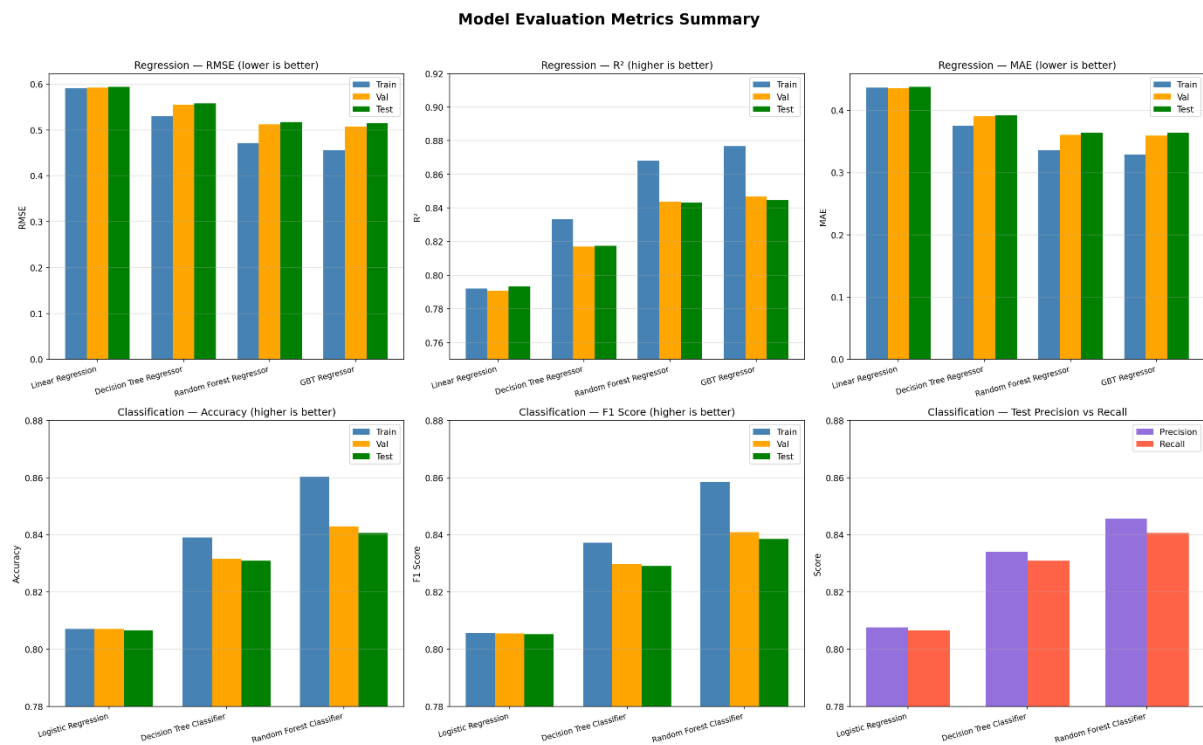


Figure 12 Model Evaluation Metrics Summary

To evaluate the predictive capabilities of our pipeline, we tracked model performance across a 70/15/15 train, validation, and test split. Figure 12 illustrates the comprehensive metric summaries for both our continuous rating predictions (Regression) and our discrete quality tier predictions (Classification).

Regression Performance (Predicting Exact IMDb Ratings): We evaluated four regression models: Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosted Trees (GBT) Regressor. The baseline Linear Regression model struggled to capture the non-linear complexities of the entertainment data, yielding the highest Test RMSE (0.5937).

Regression Modelling

Linear Regression:

Train: RMSE: 0.5917, MAE: 0.4366, R^2 : 0.7921

Val: RMSE: 0.5926, MAE: 0.4361, R^2 : 0.7907

Test: RMSE: 0.5937, MAE: 0.4376, R^2 : 0.7932

Decision Tree Regressor

Train: RMSE: 0.5304, MAE: 0.3772, R^2 : 0.8329

Val: RMSE: 0.5583, MAE: 0.3928, R^2 : 0.8142

Test: RMSE: 0.5602, MAE: 0.3968, R^2 : 0.8158

Random Forest Regressor (with CV tuning)

Train: RMSE: 0.4714, MAE: 0.3358, R^2 : 0.8680

Val: RMSE: 0.5124, MAE: 0.3605, R^2 : 0.8435

Test: RMSE: 0.5171, MAE: 0.3638, R^2 : 0.8431

Gradient Boosted Trees Regressor

Train: RMSE: 0.4557, MAE: 0.3291, R^2 : 0.8767

Val: RMSE: 0.5069, MAE: 0.3592, R^2 : 0.8468

Test: RMSE: 0.5145, MAE: 0.3639, R^2 : 0.8446

Classification Modelling

Logistic Regression

Train: Acc: 0.8071, F1: 0.8056

Val: Acc: 0.8070, F1: 0.8055

Test: Acc: 0.8066, F1: 0.8052

Decision Tree Classifier

Train: Acc: 0.8393, F1: 0.8374

Val: Acc: 0.8275, F1: 0.8255

Test: Acc: 0.8290, F1: 0.8269

Random Forest Classifier

Train: Acc: 0.8604, F1: 0.8585

Val: Acc: 0.8429, F1: 0.8409

Test: Acc: 0.8406, F1: 0.8385

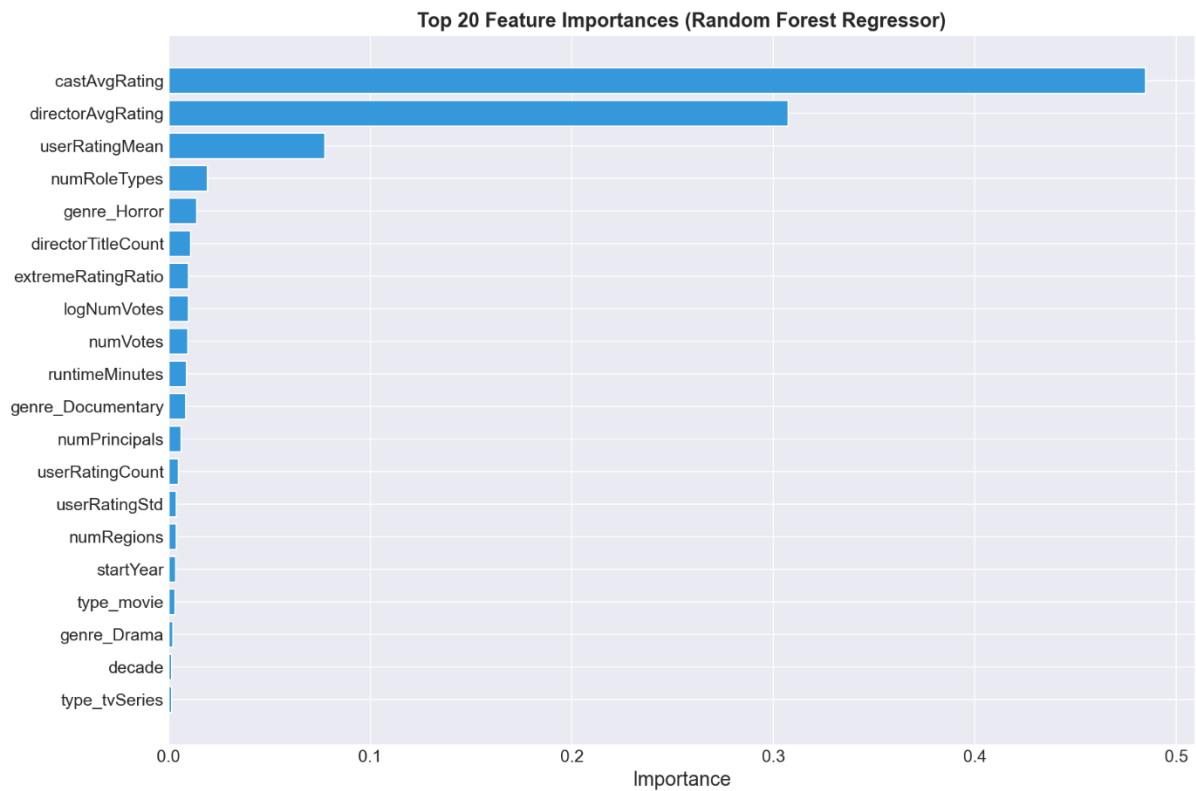


Figure 13 Random Forest Regressor Feature Importances

The figure shows the top 20 feature importances for the random forest regressor. It's clear that the cast's and director's average ratings are by far the most important features.

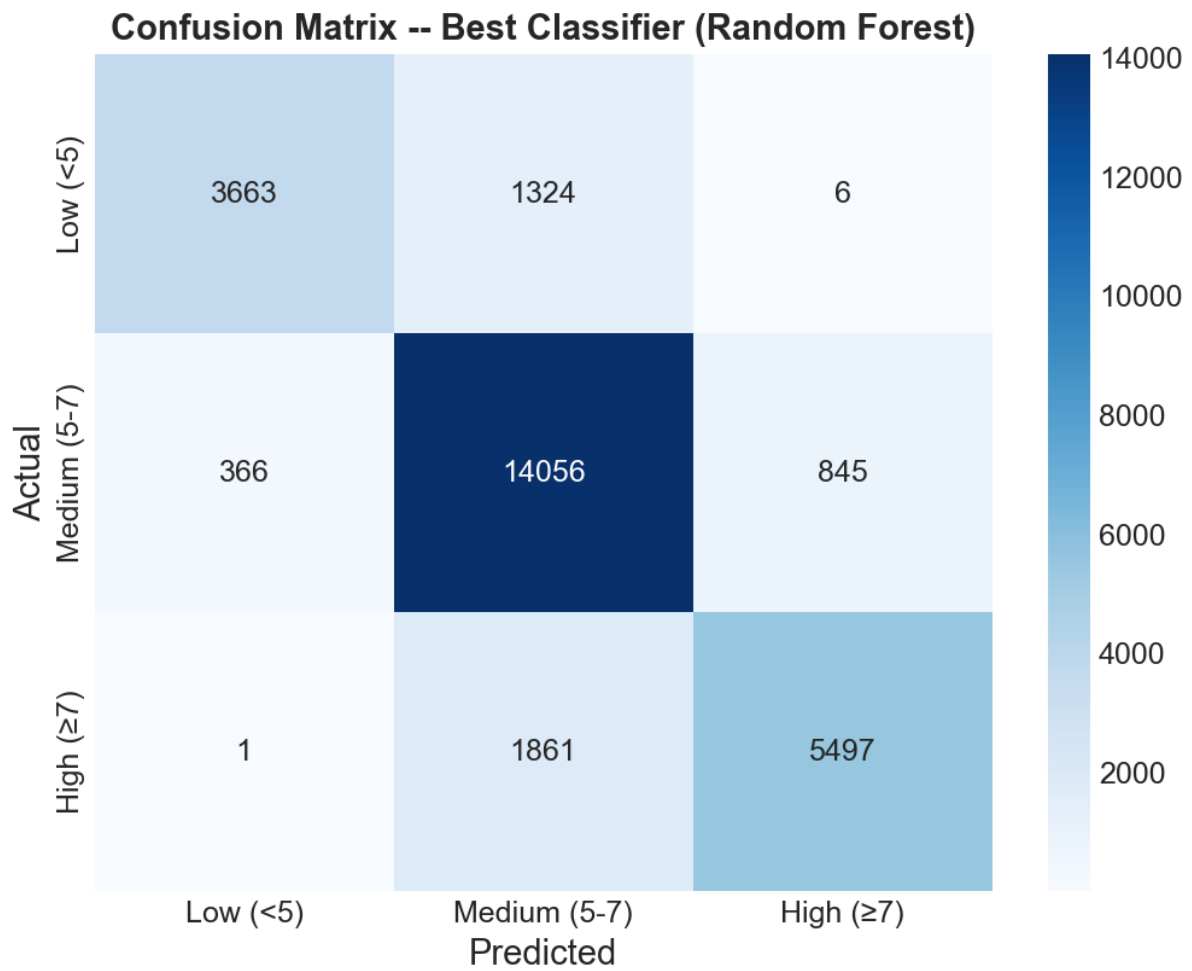


Figure 14 Random Forest Classifier Confusion Matrix

The figure above shows the confusion matrix for the random forest classifier.

6. Unsuccessful Trials

Initial attempts to perform large-scale Spark DataFrame joins during preprocessing resulted in Py4JJavaError crashes on the Python worker nodes. Memory management issues also surfaced in the iterative K-Means training phase, where successive model fits accumulated cached DataFrames in memory causing OOM errors.

Early processing iterations retained the entirety of the IMDb dataset, which included thousands of obscure titles. We determined that titles with a low number of votes had unreliable average ratings that skewed training. A threshold was applied to discard 229,123 titles with fewer than 100 votes, significantly improving data quality.

Applying K-Means clustering across the entire feature space yielded a binary segmentation ($K=2$) that was too broad to drive actionable business decisions.

7. Enhancements and Future work

The current pipeline relies primarily on structured metadata (genres, runtimes, cast history). A significant opportunity exists to integrate Natural Language Processing (NLP) techniques on title summaries and reviews to extract sentiment features.

An analysis of the current dataset indicates a significant bias, with English-language titles comprising approximately 75.5% of the data. This skew can lead to models that perform well on English content but struggle to accurately predict audience reception for other languages.

While static visualizations are currently generated during the evaluation phase, a dynamic dashboard will significantly enhance the usability of the project's outputs for Key Stakeholders.